Results

Classical Process Model

Model summary

	AIC	BIC	Log-likelihood	n	df
Model 1	4160.444	4186.496	−2070.222	100	0

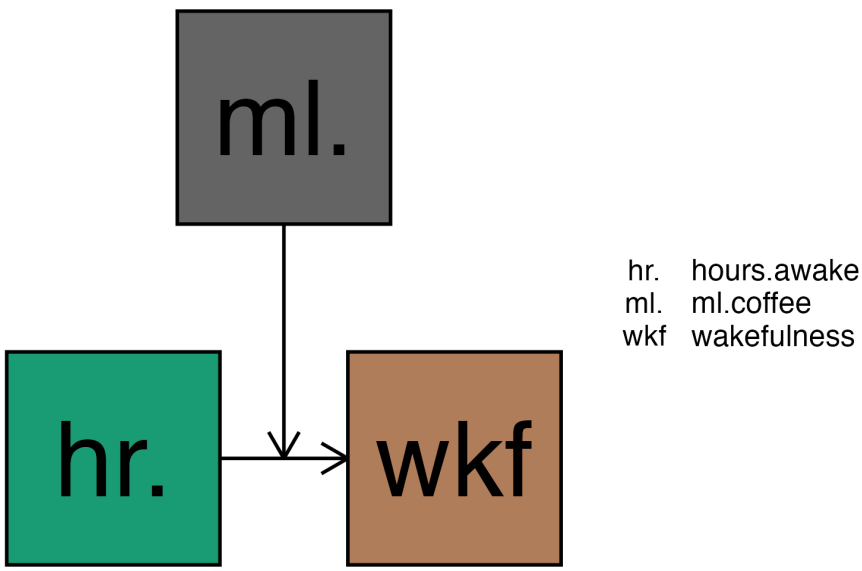
R-squared

	R^2
	Model 1
wakefulness	0.989

Path plots

Model 1

Conceptual path plot



Parameter estimates

Important: Parameter estimates can only be interpreted as causal effects if all confounding effects are accounted for and if the causal effect directions are correctly specified.

Model 1

Path coefficients

			Estimate	Std. Error	z-value	p	95% Confidence Interval Lower	95% Confidence Interval Upper
hours.awake	→	wakefulness	23.013	0.344	66.911	< .001	22.339	23.687
ml.coffee	→	wakefulness	1.269	0.020	63.595	< .001	1.230	1.309
hours.awake:ml.coffee	→	wakefulness	0.102	0.004	25.895	< .001	0.095	0.110

Note. Moderation effect estimates are based on mean-centered variables.

Direct and indirect effects

			ml.coffee	Estimate	Std. Error	z-value	p	95% Confidence Interval Lower	95% Confidence Interval Upper
hours.awake	→	wakefulness	16	13.651	0.511	26.699	< .001	12.649	14.653
hours.awake	→	wakefulness	50	23.635	0.344	68.791	< .001	22.961	24.308
hours.awake	→	wakefulness	84	32.713	0.496	65.992	< .001	31.741	33.684

Note. Moderation effect estimates are based on mean-centered variables.

Total effects

				ml.coffee	Estimate	Std. Error	z-value	p	95% Confidence Interval Lower	95% Confidence Interval Upper
Total	hours.awake	→	wakefulness	16	13.651	0.511	26.699	< .001	12.649	14.653
	hours.awake	→	wakefulness	50	23.635	0.344	68.791	< .001	22.961	24.308
	hours.awake	→	wakefulness	84	32.713	0.496	65.992	< .001	31.741	33.684

Note. Moderation effect estimates are based on mean-centered variables.

Linear Regression

Model Summary – wakefulness

Model	R	R^2	Adjusted R^2	RMSE	R^2 Change	df1	df2	p
M_0	0.000	0.000	0.000	163.406	0.000	0	99	
M_1	0.955	0.912	0.910	48.893	0.912	2	97	< .001
M_2	0.994	0.989	0.988	17.705	0.076	1	96	< .001

Note. M_1 includes hours.awake, ml.coffee

Note. M_2 includes hours.awake, ml.coffee, hours.awake:ml.coffee

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	2.412e+6	2	1.206e+6	504.412	< .001
	Residual	231876.234	97	2390.477		
	Total	2.643e+6	99			
M_2	Regression	2.613e+6	3	871118.347	2779.123	< .001
	Residual	30091.281	96	313.451		
	Total	2.643e+6	99			

Note. M_1 includes hours.awake, ml.coffee

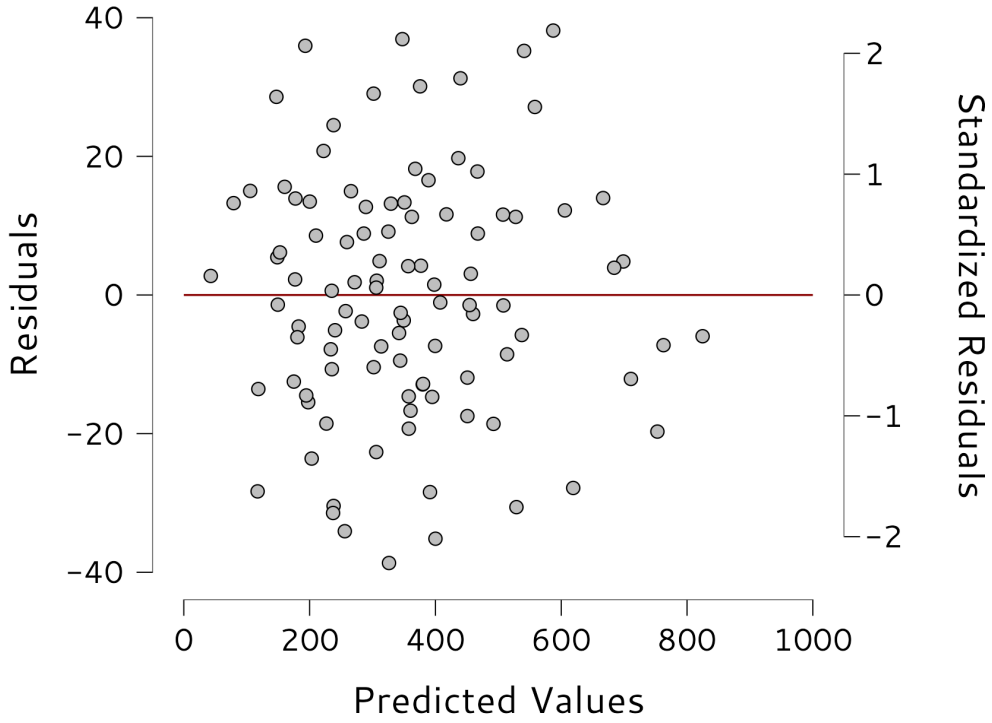
Note. M_2 includes hours.awake, ml.coffee, hours.awake:ml.coffee

Note. The intercept model is omitted, as no meaningful information can be shown.

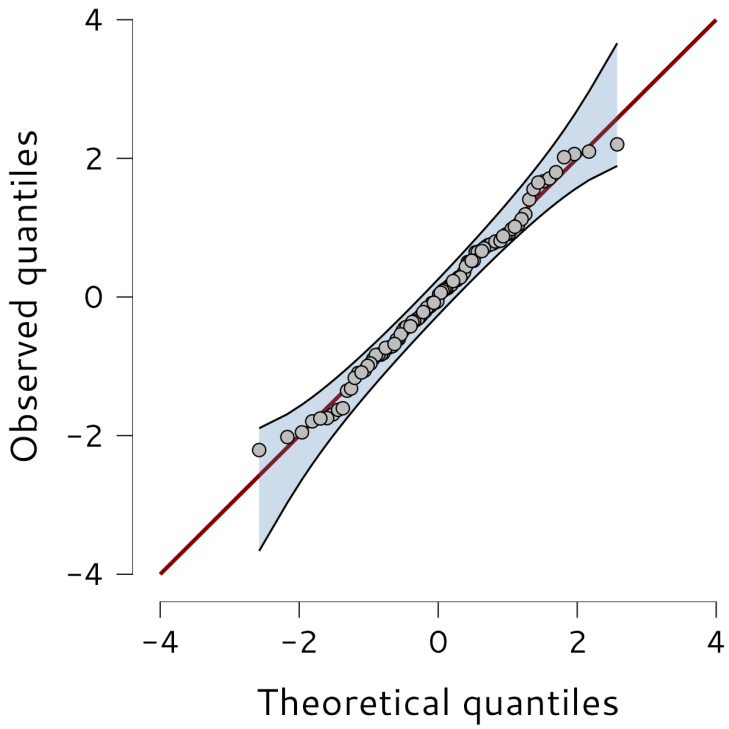
Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
M_0	(Intercept)	353.775	16.341		21.650	< .001
M_1	(Intercept)	−194.227	18.475		−10.513	< .001
	hours.awake	23.458	0.968	0.730	24.229	< .001
	ml.coffee	1.224	0.056	0.658	21.838	< .001
M_2	(Intercept)	61.683	12.103		5.096	< .001
	hours.awake	−3.612	1.123	−0.112	−3.216	.002
	ml.coffee	0.266	0.043	0.143	6.197	< .001
	hours.awake ✻ ml.coffee	0.102	0.004	1.001	25.372	< .001

Residuals vs. Predicted



Q–Q Plot Standardized Residuals



Descriptives Plots

hours.awake – wakefulness

